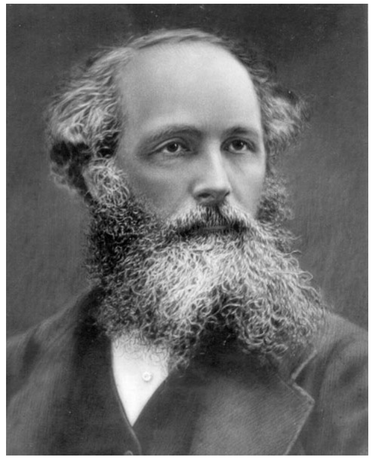
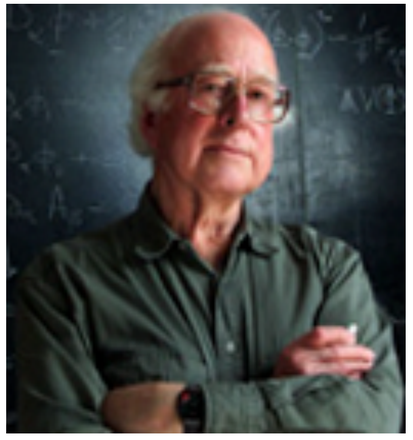
BMS strings

Quantisation and spectra of non-lorentzian bosonic strings





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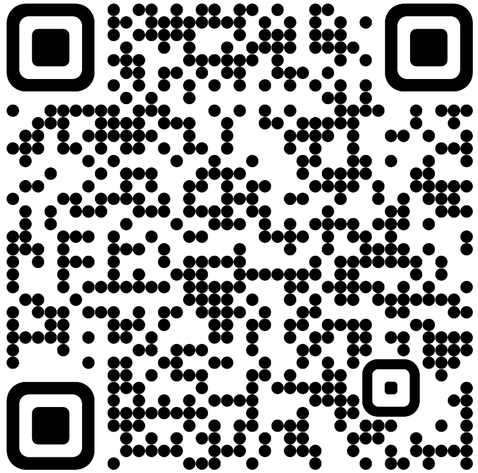
Based on the following papers:

- **arXiv:2407.12778** with **Girish Vishwa**
- **arXiv:2505.03462** with **Girish Vishwa**
- **arXiv:2509.04397** with **Emil Have** and **Niels Obers**
- **arXiv:2606.05500** with **Girish Vishwa**

and works in progress with **Ismaël Ahlouche**, **Emil Have** and **Niels Obers**

Outline

- “Non-relativistic” strings
- BMS strings
- BRST quantisation
- The spectrum of the Carroll string
- The spectrum of the bosonic ambitwistor string
- Conclusions and outlook



“Non-relativistic” strings

Non-relativistic strings

- Bosonic strings are described by a 2d sigma model with fields $X: \Sigma \to M$
  - Σ is a two-dimensional “worldsheet”
  - M is the ambient spacetime on which the string propagates
- For the traditional ***relativistic*** string, both Σ and M are ***lorentzian**** spacetimes
- For ***non-relativistic*** strings, either Σ or M (or both) are ***non-lorentzian*** spacetimes
- e.g., the **Gomis–Ooguri** string keeps Σ lorentzian but needs M to be (“stringy”) galilean

*often we “Wick rotate” Σ

Why study non-relativistic strings?

- The **ambitwistor string**, whose correlators encode **massless scattering**, has a carrollian worldsheet
- Strings near **black hole horizons** are described by a carrollian target space: the distance from the horizon becoming an “effective” speed of light
- Non-relativistic strings are part of a **web of dualities** including the usual relativistic strings
- In AdS/CFT, **near-BPS limits** of field theories often have a non-relativistic dual geometry
- Final (?) frontier of the **kinematical Erlangen Programme**

Σ determines the string gauge algebra

- String sigma models have first-class constraints, whose Poisson brackets define the gauge algebra of the string theory
- The gauge algebra is determined by the geometry of Σ:
  - (Two copies of) Virasoro if Σ is lorentzian
  - $\boldsymbol{BMS_3}$ if Σ is carrollian (***BMS = Bondi–(Van der Burg)–Metzner–Sachs)***
  - $\boldsymbol{GCA_2}$ if Σ is galilean (***GCA = Galilean Conformal Algebra)***
- Of course, in two dimensions $\boldsymbol{GCA_2 \cong BMS_3}$, and we will use the carrollian language.

String spectra

- Strings can be quantised à la BRST: the physical spectrum is given by the BRST cohomology
- Technically, it is a ***semi-infinite cohomology relative to the centre*** in the sense of **Feigin**
- This is a sort of Lie algebra cohomology and as such it requires both a **Lie algebra** and a **representation**
- Σ determines the Lie algebra
- M determines the representation
- e.g., the Gomis–Ooguri string is a Virasoro string, but the representation is different from the usual bosonic string: in the holomorphic sector, $T_{\text{bos}} = -\frac{1}{2}\eta_{\mu\nu}\partial X^\mu \partial X^\nu$ whereas $T_{\text{GO}} = -\frac{1}{2}\underbrace{\partial X^{\mathsf{T}} \cdot \partial X^{\mathsf{T}}}_{\text{transverse}} - \beta\partial\gamma$

BMS strings

The $\boldsymbol{BMS_3}$ algebra

- The centreless $\boldsymbol{BMS_3}$ algebra has generators $\{L_n, M_n\}_{n\in\mathbb{Z}}$ obeying the following Lie bracket

$$[L_n, L_m] = (n-m)L_{n+m} \qquad [L_n, M_m] = (n-m)M_{n+m} \qquad [M_n, M_m] = 0$$

- This Lie algebra admits a two-parameter family of non-trivial central extensions:

$$[L_n, L_m] = (n-m)L_{n+m} + \frac{1}{12}c_L n(n^2-1)\delta_{n+m,0}$$

$$[L_n, M_m] = (n-m)M_{n+m} + \frac{1}{12}c_M n(n^2-1)\delta_{n+m,0}$$

$$[M_n, M_m] = 0$$

OPE formalism

- It is convenient for calculations to introduce a **formal** variable z (no geometric interpretation is implied!)

  and fields $T(z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}$ and $M(z) = \sum_{n\in\mathbb{Z}} M_n z^{-n-2}$

- The $\boldsymbol{BMS_3}$ algebra of modes is equivalent to the operator product expansions

$$T(z)\,T(w) = \frac{\frac{1}{2}c_L\mathbf{1}}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \text{reg.}$$

$$T(z)\,M(w) = \frac{\frac{1}{2}c_M\mathbf{1}}{(z-w)^4} + \frac{2M(w)}{(z-w)^2} + \frac{\partial M(w)}{z-w} + \text{reg.}$$

$$M(z)\,M(w) = \text{reg.}$$

BRST complexes for $\boldsymbol{BMS_3}$

[JMF+Vishwa, 2407.12778]

- The $\boldsymbol{BMS_3}$ algebra admits a BRST complex **if and only if** the representation (T, M) has central charges $(c_L, c_M) = (52,0)$

- The BRST differential d is the zero mode of the BRST current

$$j_{\text{BRST}} = (cT) + (CM) + \frac{1}{2}(cT^{\text{gh}}) + \frac{1}{2}(CM^{\text{gh}})$$

where (b, c) and (B, C) are fermionic BC systems of weights $(2, -1)$ and

$$T^{\text{gh}} = -2b\partial c - \partial bc - 2B\partial C - \partial BC \quad \text{and} \quad M^{\text{gh}} = -2B\partial c - \partial Bc$$

A general class of examples

There are a number of relevant representations (T, M) of $\boldsymbol{BMS_3}$ with $(c_L, c_M) = (52,0)$ which can be constructed by introducing 26 bosonic BC systems (X^μ, Π_μ) of weights $(0,1)$

The defining OPEs are

$$\Pi_\mu(z)\, X^\nu(w) = \frac{\delta_\mu^\nu \mathbf{1}}{z-w} + \text{reg.} \qquad \Pi_\mu(z)\, \Pi_\nu(w) = \text{reg.} \qquad X^\mu(z)\, X^\nu(w) = \text{reg.}$$

Introduce the modes $X^\mu(z) = \sum_{n\in\mathbb{Z}} x_n^\mu\, z^{-n}$ and $\Pi_\mu(z) = \sum_{n\in\mathbb{Z}} (\pi_\mu)_n\, z^{-n-1}$

and the OPEs are equivalent to the mode algebra

$$[(\pi_\mu)_n, x_m^\nu] = \delta_\mu^\nu \delta_{n+m,0} \qquad [(\pi_\mu)_n, (\pi_\nu)_m] = 0 \qquad [x_n^\mu, x_m^\nu] = 0$$

A general class of examples (II)

In all cases we have $T = \partial X^\mu \, \Pi_{\mu'}$ but M varies:

- the **ambitwistor string**: $M = -\frac{1}{2}\eta^{\mu\nu}\Pi_\mu \, \Pi_\nu$ with $\eta^{\mu\nu}$ the $d = 26$ Minkowski metric

[Mason+Skinner]

- the **Carroll string**: $M = \frac{1}{2}\Pi_0^2 - \frac{1}{2}\delta_{ij}\partial X^i \, \partial X^j$

[Blair+Lahnsteiner+Obers+Yan, Gomis+Yan, Bagchi+Banerjee+Hartong+Have+Kolekar]

- the Galilei string: $M = \frac{1}{2}(\partial X^0)^2 - \frac{1}{2}\delta^{ij}\Pi_i \, \Pi_j$

[Batlle+Gomis+Not]

A general class of examples (III)

[JMF+Have+Obers, work in progress]

There are also more exotic strings:

- "p-brane carrollian" strings: $M = -\frac{1}{2}\eta^{\alpha\beta}\Pi_\alpha \Pi_\beta - \frac{1}{2}\delta_{ab}\partial X^a \partial X^b$ with $\alpha, \beta \in \{0,1,\ldots,p\}$ and $a, b \in \{p+1,\ldots,25\}$
- "instantonic" strings: $M = -\frac{1}{2}\eta_{\mu\nu}\partial X^\mu \partial X^\nu$ (galilean) or $M = -\frac{1}{2}\delta_{\mu\nu}\partial X^\mu \partial X^\nu$ (carrollian)
- "p-brane galilean" strings: $M = -\frac{1}{2}\eta_{\alpha\beta}\partial X^\alpha \partial X^\beta - \frac{1}{2}\delta^{ab}\Pi_a \Pi_b$ with $\alpha, \beta \in \{0,1,\ldots,p\}$ and $a, b \in \{p+1,\ldots,25\}$

BRST quantisation

The Carroll string

- (X^μ, Π_μ) 26 bosonic BC systems with weights $(0,1)$

- $T = \partial X^\mu \Pi_\mu$ and $M = \frac{1}{2}\Pi_0^2 - \frac{1}{2}\delta_{ij}\partial X^i \partial X^j$

$$T(z)T(w) = \frac{26\ \mathbf{1}}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \text{reg.}$$

$$T(z)M(w) = \frac{2M(w)}{(z-w)^2} + \frac{\partial M(w)}{z-w} + \text{reg.}$$

$$M(z)M(w) = \text{reg.}$$

Bosonic ambitwistor string

- (X^μ, Π_μ) 26 bosonic BC systems with weights $(0,1)$

- $T = \partial X^\mu \Pi_\mu$ and $M = -\frac{1}{2}\eta^{\mu\nu}\Pi_\mu\Pi_\nu$

$$T(z)T(w) = \frac{26\ \mathbf{1}}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \text{reg.}$$

$$T(z)M(w) = \frac{2M(w)}{(z-w)^2} + \frac{\partial M(w)}{z-w} + \text{reg.}$$

$$M(z)M(w) = \text{reg.}$$

BRST complex

Vacuum and mode expansions

- Mode expansions

$$X^\mu(z) = \sum_{n\in\mathbb{Z}} x^\mu_n z^{-n} \qquad \Pi_\mu(z) = \sum_{n\in\mathbb{Z}} (\pi_\mu)_n z^{-n-1}$$

$$b(z) = \sum_{n\in\mathbb{Z}} b_n z^{-n-2} \qquad c(z) = \sum_{n\in\mathbb{Z}} c_n z^{-n+1}$$

$$B(z) = \sum_{n\in\mathbb{Z}} B_n z^{-n-2} \qquad C(z) = \sum_{n\in\mathbb{Z}} C_n z^{-n+1}$$

- The vacuum $|0\rangle$ is annihilated by $c_{n\geq 2}, C_{n\geq 2}, x^\mu_{n\geq 1}, (\pi_\mu)_{n\geq 0}, b_{n\geq -1}, B_{n\geq -1}$

BRST complex (II)

Cochains

- Bigraded by **ghost number**

$$\mathrm{gh}(B) = \mathrm{gh}(b) = -1 \qquad \mathrm{gh}(C) = \mathrm{gh}(c) = 1 \qquad \mathrm{gh}(X^\mu) = \mathrm{gh}(\Pi_\mu) = 0$$

- and by **conformal weight**: the eigenvalue of $L_0^{\mathrm{tot}} = L_0 + L_0^{\mathrm{gh}}$

$$T = \sum_{n\in\mathbb{Z}} L_n z^{-n-2} \qquad T^{\mathrm{gh}} = \sum_{n\in\mathbb{Z}} L_n^{\mathrm{gh}} z^{-n-2}$$

- $L_0^{\mathrm{tot}} = [d, b_0]$ acts diagonally and trivially on cohomology, so $\mathsf{C}^\bullet = \ker L_0^{\mathrm{tot}}$ is a quasi-isomorphic subcomplex, graded by ghost number

BRST complex (III)

Momentum space cochains

- j_{BRST} only depends on ∂X^μ, so the zero mode x_0^μ does not appear
- Therefore in momentum space, the BRST differential is algebraic

$$|p\rangle = e^{ip\cdot x}\,|0\rangle \qquad\qquad (\pi_\mu)_0\,|p\rangle = ip_\mu\,|p\rangle$$

- So we must complexify and for each p we have a complex $\mathsf{C}^\bullet(p)$ graded by ghost number and such that complex conjugation relates $\mathsf{C}^\bullet(p)$ and $\mathsf{C}^\bullet(-p)$
- It is a finite complex: $\mathsf{C}^n(p) \neq 0$ only for $n = 0,1,\ldots,6$

BRST complex (IV)

Relative subcomplex

- The relative subcomplex $\mathsf{C}^\bullet_{\mathrm{rel}}(p) = \mathsf{C}^\bullet(p) \cap \ker b_0$, with $\mathsf{C}^n_{\mathrm{rel}}(p) \neq 0$ only for $n = 0,1,\ldots,5$

$$0 \longrightarrow \mathsf{C}^n_{\mathrm{rel}}(p) \xrightarrow{\ i\ } \mathsf{C}^n(p) \xrightarrow{\ b_0\ } \mathsf{C}^{n-1}_{\mathrm{rel}}(p) \longrightarrow 0$$

$$\mathsf{C}^{n-1}_{\mathrm{rel}}(p) \xrightarrow{\ c_0\ } \mathsf{C}^n(p)$$

- inducing a long exact sequence in cohomology

$$\cdots \longrightarrow \mathsf{H}^{n-2}_{\mathrm{rel}}(p) \xrightarrow{\ \Delta\ } \mathsf{H}^n_{\mathrm{rel}}(p) \xrightarrow{\ i_*\ } \mathsf{H}^n(p) \xrightarrow{\ (b_0)_*\ } \mathsf{H}^{n-1}_{\mathrm{rel}}(p) \xrightarrow{\ \Delta\ } \mathsf{H}^{n+1}_{\mathrm{rel}}(p) \longrightarrow \cdots$$

where the connecting homomorphism $\Delta = [d, c_0]$

The spectrum of the Carroll string

Structural results

- $\mathrm{H}^\bullet(p) = 0$ unless $p_0 = 0$

$\Rightarrow$ the spectrum consists of states with zero energy!

(The energy is a Casimir of the Carroll algebra.)

- Let $p = (0, \boldsymbol{p})$. The BRST cohomology $\mathrm{H}^\bullet(p)$ is naturally a representation of the stabiliser $\mathrm{SO}(\boldsymbol{p}^\perp)$ of $\boldsymbol{p}$ in $\mathrm{SO}(25)$
- If $\boldsymbol{p} \neq \boldsymbol{0}$, then $\mathrm{SO}(\boldsymbol{p}^\perp) \cong \mathrm{SO}(24)$ is compact and acts fully reducibly on the complex and on the cohomology
- If $\boldsymbol{p} = \boldsymbol{0}$, then $\mathrm{SO}(\boldsymbol{p}^\perp) \cong \mathrm{SO}(25)$ is also compact and acts fully reducibly
- The BRST cohomology exhibits Poincaré duality $\mathrm{H}^n(p) \cong \mathrm{H}^{6-n}(p)$

The spectrum of the Carroll string

(for $\boldsymbol{p} \neq \mathbf{0}$)

$$\mathrm{H}^0(0,\boldsymbol{p}) \cong \mathrm{H}^6(0,\boldsymbol{p}) = 0$$

$$\mathrm{H}^1(0,\boldsymbol{p}) \cong \mathrm{H}^5(0,\boldsymbol{p}) \cong \mathbb{C} \oplus V^\perp$$

$$\mathrm{H}^2(0,\boldsymbol{p}) \cong \mathrm{H}^4(0,\boldsymbol{p}) \cong 4\mathbb{C} \oplus 4V^\perp \oplus \odot_0^2 V^\perp \oplus \wedge^2 V^\perp$$

$$\mathrm{H}^3(0,\boldsymbol{p}) \cong 6\mathbb{C} \oplus 6V^\perp \oplus 2\,\odot_0^2 V^\perp \oplus 2\,\wedge^2 V^\perp$$

where the representations $\mathbb{C}$, $V^\perp = \boldsymbol{p}^\perp$, $\odot_0^2 V^\perp$ and $\wedge^2 V^\perp$ are the (complexified) scalar, vector, symmetric traceless and skew-symmetric tensor representations of $\mathrm{SO}(\boldsymbol{p}^\perp) \cong \mathrm{SO}(24)$

There is enhancement of cohomology for $\boldsymbol{p} = \mathbf{0}$, just as in the usual bosonic string

Physical interpretation

(in terms of UIRs of the Carroll group)

- $\mathsf{H}^\bullet(0,\boldsymbol{p})$ can be interpreted as inducing representations for **unitary irreducible representations** (UIRs) of the Carroll group, constructed by Mackey's method of induced representations (cf. Wigner for Poincaré)

- These UIRs are infinite-dimensional and are carried by square-integrable sections of homogeneous vector bundles

$$\mathrm{SO}(25) \times_{\mathrm{SO}(\boldsymbol{p}^\perp)} \mathsf{H}^\bullet(0,\boldsymbol{p}) \to S^{24}(\|\boldsymbol{p}\|)$$

- The UIRs of the four-dimensional Carroll group were classified in collaboration with **Alfredo Pérez** and **Stefan Prohazka** and the UIRs appearing in the Carroll string are the 26-dimensional analogues of what we termed the ***aristotelion*** UIRs.

[JMF+Pérez+Prohazka, 2307.05674]

Deforming the target background

- A carrollian structure with Kalb–Ramond field is a triple $(\kappa^\mu, h_{\mu\nu}, B_{\mu\nu})$ with $\kappa^\mu h_{\mu\nu} = 0$
- For Carroll spacetime $\kappa^\mu = \delta^\mu_0$, $h_{\mu\nu} = \delta_{ij}\delta^i_\mu\delta^j_\nu$ and $B_{\mu\nu} = 0$
- Based on our intuition with the closed bosonic string, we expect the spectrum to contain first-order deformations of the carrollian structure of Carroll spacetime:

$$\kappa^\mu = \delta^\mu_0 + \delta\kappa^\mu \qquad h_{\mu\nu} = \delta_{ij}\delta^i_\mu\delta^j_\nu + \delta h_{\mu\nu} \qquad B_{\mu\nu} = 0 + \delta B_{\mu\nu}$$

and the condition $\kappa^\mu h_{\mu\nu} = 0$ becomes to first order: $\delta h_{00} = 0$ and $\delta h_{0i} = -\delta_{ij}\delta\kappa^j$

First-order deformations

(as complex representations of $\mathbf{SO}(\boldsymbol{p}^\perp)$)

For $p = (0, \boldsymbol{p})$ and $\boldsymbol{p} \neq \mathbf{0}$

$\delta\kappa^0$	$\mathbb{C}$
$\delta\kappa^i$	$\mathbb{C} \oplus V^\perp$
δh_{ij}	$2\mathbb{C} \oplus V^\perp \oplus \odot_0^2 V^\perp$
δB_{ij}	$V^\perp \oplus \wedge^2 V^\perp$
δB_{0i}	$\mathbb{C} \oplus V^\perp$

Not all these deformations are physical. We have diffeomorphisms and gauge transformations of the Kalb–Ramond field: $\delta\kappa = \mathscr{L}_\zeta \kappa$ $\quad \delta h = \mathscr{L}_\zeta h$ and $\delta B = \mathscr{L}_\zeta B + d\theta$

where $\zeta = \zeta^\mu \partial_\mu$ and $\theta = \theta_\mu dx^\mu$

First-order deformations (II)

(as representations)

- For Carroll spacetime and in momentum space

$$\delta\kappa^0 = \delta\kappa^i = \delta h_{00} = \delta h_{0i} = 0 \qquad \delta h_{ij} = 2ip_{(i}\zeta_{j)} \qquad \delta B_{0i} = -i\theta_0 p_i \qquad \delta B_{ij} = 2ip_{[i}\theta_{j]}$$

- Notice that only the transverse part $\theta_i^\perp$ of θ_i contributes to δB_{ij}, so the effective gauge parameters are $\zeta^i, \theta_0, \theta_i^\perp$ or in terms of (complex) representations $2\mathbb{C} \oplus 2V^\perp$

- Modulo gauge transformations, $\mathrm{H}^2(0, \boldsymbol{p})$ contains a subspace isomorphic to $3\mathbb{C} \oplus 2V^\perp \oplus \odot_0^2 V^\perp \oplus \wedge^2 V^\perp$, which we may try to put into correspondence with the first-order deformations of the geometry

- and a subspace isomorphic to $\mathbb{C} \oplus 2V^\perp$ which resists (thus far) a geometric interpretation

Vertex operators

(for geometric deformations in $\mathsf{H}^2(0,\boldsymbol{p})$)

- δh_{ij} : $cC\left(\partial X^i \partial X^j + \delta^{ij}\Pi_0^2\right) e^{ip\cdot X}$ (carrollian "graviton")
- δB_{ij} : $cC\left(\partial X^{[i}\Pi^{j]} + \frac{i}{2}p^{[i}\partial^2 X^{j]}\right) e^{ip\cdot X}$ (carrollian "B-field")
- $\delta h_{0i} = -\delta_{ij}\delta\kappa^j$: $cC\left(\partial X^i \partial X^0 + \Pi^i\Pi_0 - \frac{i}{2}p^i\partial\Pi_0\right) e^{ip\cdot X}$
- δB_{0i} : $cC\partial X^i \Pi_0 e^{ip\cdot X}$

- Every class of the form $\boldsymbol{cCW}$ in $\mathsf{H}^2(0,\boldsymbol{p})$, with W not involving ghosts, appears as $\partial ccCW$ and $\partial CcCW$ in $\mathsf{H}^3(0,\boldsymbol{p})$ and as $\partial c\partial CcCW$ in $\mathsf{H}^4(0,\boldsymbol{p})$
- The transverse trace $\delta^{ij}\delta h_{ij}$ hides the first-order deformation $\delta\kappa^0$ of the time component of $\boldsymbol{\kappa}$

The spectrum of the bosonic ambitwistor string

Some general results

- $\mathrm{H}^\bullet_{\mathrm{rel}}(p) = 0$ unless $p^2 := \eta^{\mu\nu} p_\mu p_\nu = 0$
- The long exact cohomology sequence $\Rightarrow \mathrm{H}^\bullet(p) = 0$ unless $p^2 = 0$
- $\mathrm{H}^\bullet(p)$ and $\mathrm{H}^\bullet_{\mathrm{rel}}(p)$ are modules over $\mathrm{Stab}(p) \subset \mathrm{SO}(25,1)$
- If $p = 0$ then $\mathrm{Stab}(p) = \mathrm{SO}(25,1)$, being semisimple, acts fully reducibly
- If $p \neq 0$ but $p^2 = 0$, then $\mathrm{Stab}(p)$ does **not** act fully reducibly

Relative cohomology

$p^2 = 0$ but $p \neq 0$

- $\mathsf{H}^n_{\text{rel}}(p) = 0$ except for $n = 2,3$
- $\mathsf{H}^2_{\text{rel}}(p)$ represented by

$$G_{\mu\nu}(p) cC \left(\Pi^\mu \Pi^\nu - i p^{(\mu} \partial \Pi^{\nu)} \right) e^{ip\cdot X} + F_{\mu\nu}(p) cC \partial X^\mu \Pi^\nu e^{ip\cdot X}$$

where $G_{\mu\nu} = G_{\nu\mu}$ and $F_{\mu\nu} = -F_{\nu\mu}$ subject to the **cocycle** conditions

$$p^\mu p^\nu G_{\mu\nu} = 0 \qquad\qquad p^\mu F_{\mu\nu} = 0$$

modulo **coboundaries**

$$\delta G_{\mu\nu} = i p_{(\mu} \chi_{\nu)} - \tfrac{i}{6} \eta_{\mu\nu} p\cdot\chi \qquad\qquad \delta F_{\mu\nu} = i p_{[\mu} \omega_{\nu]} \qquad\qquad p\cdot\omega = 0$$

Relative cohomology (II)

$p^2 = 0$ but $p \neq 0$

- $\mathsf{H}^3_{\text{rel}}(p)$ represented by

$$G_{\mu\nu}(p) cC\partial C\partial X^\mu \partial X^\nu e^{ip\cdot X} - \tfrac{1}{20}\,\mathrm{tr}\, G\left(cC\partial C\partial X\cdot\partial X - C\partial C\partial^2 C\right) e^{ip\cdot X} + F_{\mu\nu}(p) cC\partial C\partial X^\mu \Pi^\nu e^{ip\cdot X}$$

where $G_{\mu\nu} = G_{\nu\mu}$ and $F_{\mu\nu} = -F_{\nu\mu}$ subject to the **cocycle** conditions

$$p^\mu G_{\mu\nu} = 0 \qquad p^\mu F_{\mu\nu} = 0$$

modulo **coboundaries**

$$\delta G_{\mu\nu} = p_\mu p_\nu \theta \qquad \delta F_{\mu\nu} = i p_{[\mu}\omega_{\nu]} \qquad p\cdot\omega = 0$$

Some representations of $\mathbf{Stab}(p)$

$p^2 = 0$ but $p \neq 0$

- Let V be the vector representation of $\mathbf{SO}(25,1)$. We have musical isomorphisms induced by η

$$V \underset{\sharp}{\overset{\flat}{\rightleftarrows}} V^*$$

- $p \in V^*$ and $p^\sharp \in V$

- As a representation of $\mathbf{Stab}(p)$, the vector representation V is not irreducible but indecomposable:

$$\ell_p \subset p^\circ \subset V$$

where $\ell_p = \mathbb{R}p^\sharp$ and $p^\circ = (p^\sharp)^\perp$

Complexifying

- Let $\mathbb{V} = \mathbb{C} \otimes_{\mathbb{R}} V$ be the complexified vector representation of $\mathbf{SO(25,1)}$
- We extend η complex-linearly and we have complex-linear musical isomorphisms

$$\mathbb{V} \underset{\sharp}{\overset{\flat}{\rightleftarrows}} \mathbb{V}^*$$

- As representations of $\mathbf{Stab}(p)$

$$L_p \subset p^\perp \subset \mathbb{V}$$

where $L_p = \mathbb{C}p^\sharp$ and now $p^\perp = \mathbb{C} \otimes_{\mathbb{R}} p^\circ$

The cohomology as representations

$\mathsf{H}^2(p)$

- $\mathsf{H}^2(p) \cong \mathsf{H}^2_{\mathrm{rel}}(p)$ is **not** fully reducible. There is a non-split short exact sequence:

$$0 \longrightarrow \left(p^\perp/L_p\right)^{\otimes 2} \longrightarrow \mathsf{H}^2_{\mathrm{rel}}(p) \longrightarrow p^\perp/L_p \longrightarrow 0$$

- $\mathsf{H}^2(p)$ is **not** unitary [cf. Berkovits+Lize]

- the maximal unitary subrepresentation is

$$\left(p^\perp/L_p\right)^{\otimes 2} \cong \odot^2_0(p^\perp/L_p) \oplus \mathbb{C} \oplus \wedge^2(p^\perp/L_p)$$

corresponding to the (complexified) **massless** sector of the **closed bosonic string**

but the cohomology is strictly **larger**!

The cohomology as representations (II)

$\mathsf{H}^3(p)$ and $\mathsf{H}^4(p)$

- As representations of $\mathbf{Stab}(p)$

$$\mathsf{H}^4(p) \cong \mathsf{H}^3_{\mathrm{rel}}(p) \cong \left(\mathsf{H}^2_{\mathrm{rel}}(p)\right)^*$$

- and $\mathsf{H}^3(p)$ is an extension of $\mathsf{H}^2(p)$ by $\mathsf{H}^4(p) \cong \mathsf{H}^2(p)^*$

$$0 \longrightarrow \mathsf{H}^4(p) \longrightarrow \mathsf{H}^3(p) \longrightarrow \mathsf{H}^2(p) \longrightarrow 0$$

- **Open question**: is the extension trivial?

Conclusions and outlook

Conclusions

- Strings with non-lorentzian worldsheets have $\boldsymbol{BMS_3}$ as the “gauge algebra”
- They can be quantised à la BRST and the spectrum is thus the semi-infinite cohomology of $\boldsymbol{BMS_3}$ relative to the centre with values in representations given by the target spacetime
- The Carroll string spectrum (with target the 26-dimensional Carroll spacetime) is concentrated at **zero energy** with the cohomology interpreted as “aristotelion” UIRs of the 26-dimensional Carroll group
- The ambitwistor string spectrum (with a Poincaré-invariant target space) is **not unitary** and is **strictly larger** than what was found earlier in the literature

Ongoing and future work

- Spectrum of the p-brane carrollian/galilean strings [work in progress with **Emil Have** and **Niels Obers**]
- Construction of the string sigma models by the method of “non-linear realisations” [work in progress with **Ismaël Ahlouche**]
- Superstrings [future work?]